Option / Maybe

Parse positive port without null

Season 6 · Functional · Intermediate

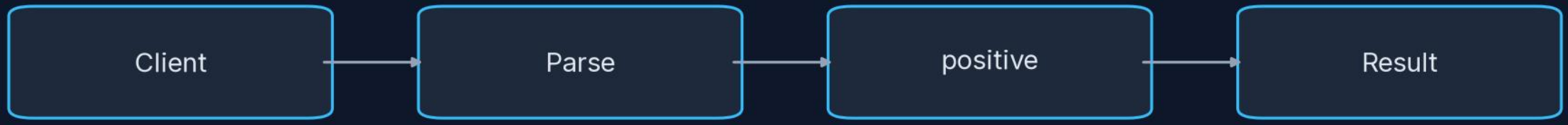
Today's Hook

- Problem: Parse positive port without null
- Naive pain cue: missing
- We will refactor to Option with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Parse positive port without null
- Constraints: solve «Parse positive port without null» with Option — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Parse positive port without null
- Success looks like: 8080

Structure



Option: Parse positive port without null

Real-World Scenario

- Running example: Parse positive port without null
- Key types: Port
- Happy-path stdout cue: 8080
- Failure-path stdout cue: missing

Naive Code

```
package com.buildwithwaqar.patterns.functional.option.problem;

import java.util.*;

public final class NaivePort {
  public static Integer parse(String s){ try { int p=Integer.parseInt(s); return p>0?p:null; } catch(Exception e){ return null; } }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Parse positive port without null
- Without Option, call sites absorb the wrong details.
- Failure mode to show on camera: missing
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.functional.option.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.concurrent.locks.*;
import java.util.function.*;

final class Port {
  static OptionalInt parse(String s){
    try { int p=Integer.parseInt(s); return p>0? OptionalInt.of(p): OptionalInt.empty(); }
    catch(NumberFormatException e){ return OptionalInt.empty(); }
  }
  static String describe(OptionalInt p){ return p.isPresent()? "port="+p.getAsInt(): "missing"; }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.functional.option.problem;
import java.util.*; import java.util.concurrent.*; import java.util.concurrent.atomic.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Option: parse positive port ===");
    System.out.println("--- naive ---"); System.out.println(NaivePort.parse("8080")); System.out.println(NaivePort.parse("-1"));
System.out.println("--- option ---");
System.out.println(Port.describe(Port.parse("8080")));
System.out.println(Port.describe(Port.parse("nope")));
System.out.println(Port.describe(Port.parse("0")));
  }
}
```

Walkthrough Output

```
=== Option: parse positive port ===
--- naive ---
8080
null
--- option ---
port=8080
missing
missing
```

Step-by-Step Trace

- Match stdout to stages — look for: 8080
- Then the injected failure: missing
- Re-run: java -cp ... 42-option ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

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Trade-offs

- Use Option when «Parse positive port without null» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Option in product language.

Related Patterns

- Name the pattern usually confused with Option and separate the forces.
- Sketch the roles for «Parse positive port without null» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Parse positive port without null
- Naive pain → Option boundary → Demo stdout.
- Remember the failure cue: missing
- Break a ProblemTest assertion, watch it fail, restore it.